

Assignment 3 – Four Phase Handshake

Please compile your solution into a single PDF file for submission.

Problem

Design a module called **fourPhaseHandshake** that transfers data from one clock domain to another. The input **dataIn** will be accepted by the module when a **validIn** input signal is asserted. Note that the module must have a **ready** output signal synchronous to **clk1** to indicate that the module is ready to accept data. As soon as data is accepted, the **ready** output signal will go low until the data has been successfully transferred to the **clk2** domain, then the **ready** output signal can be re-asserted indicating that the module is ready to accept the next data. The **validIn** signal is ignored when the **ready** output is low. The **dataIn**, **validIn** and **ready** signals are all synchronous to the **clk1** clock signal.

The module will have a **dataOut** and **validOut** signals synchronous to the **clk2** clock signal. The **validOut** is asserted for one clock cycle when a new **dataOut** byte is available on the output.

Note that the size of the data (ie 8 bits, 16 bits, 32 bits, ...) will be a parameter of the module.

Write a testbench to demonstrate the transfer of data from one clock domain to another. Include the internal **req** and **ack** signals in your waveform submitted as part of the assignment report.

In your report, include your formatted source code for the module and testbench, RTL netlist for the module, simulation waveforms and compilation report.